

AI Engineering Laboratory Manual

Lab Session 3

End-to-End Application Integrating Generative AI and Agents

Student Edition

Lab Session 3: End-to-End Application Integrating Generative AI and Agents

Context and lab intent

This lab is designed for final-year engineering students in an AI-focused program that values hands-on work, project-based learning, modern tools, and industry relevance.[page:1] It connects the earlier labs into one end-to-end application where students combine ChatGPT-style generative AI, API access, and local agent workflows built with n8n and a locally hosted Llama model.[web:22][web:58]

Lab overview

- Lab block: Sessions 7-9
- Suggested duration: 3 hours per session, 9 hours total
- Theme: End-to-end AI application with generation, agent routing, and workflow automation
- Runtime approach: local-first with optional cloud API augmentation
- Core tools: ChatGPT, Groq or compatible API, Docker, n8n, and local Llama runtime
- Pedagogy: architecture planning, implementation, integration testing, and demo presentation

Learning objectives

By the end of this lab, students should be able to:

- Design an AI application that combines chat-based generation and agent-based automation.
- Split a user problem into generation, routing, processing, and response phases.
- Integrate a local agent workflow with an API-based LLM component where appropriate.
- Build an end-to-end prototype that accepts input, reasons over it, and returns a useful result.
- Use structured outputs so that one AI component can feed the next reliably.
- Test a workflow for correctness, error handling, and user experience.
- Present the architecture and tradeoffs of the system clearly.

Prerequisites

Students should already have completed Labs 1 and 2 and should have:

- A working chat-based GenAI prototype.

- A working local n8n + Llama environment.
- Basic comfort with prompt design and API calls.
- A browser, code editor, and local runtime stack.

Suggested end-to-end project themes

Students may choose one project theme for the lab series:

- AI study assistant.
- Resume and interview helper.
- College support desk bot.
- Assignment summarizer and planner.
- Personalized learning assistant.
- Simple research assistant for academic notes.

Session structure

Session	Focus	Core outcome
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7	Architecture and data flow design	Students define the full AI system before building
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8	Implementation of integrated workflow	Students build the app and connect components
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9	Testing, refinement, and demo	Students harden the app and present results
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Session 7

Architecture and data flow design

Session objective

Students should learn how to plan an AI application before building it, identifying where generation, routing, and automation should happen.

Concept coverage

- Frontend input versus backend orchestration
- Difference between one-shot LLM use and multi-step agent flow
- When to use local models versus API-based models
- Structured handoff between workflow stages
- Data formats, prompts, state, and validation

Instructor demonstration

The instructor should show a sample architecture:

1. User enters a request.
2. A generation step rewrites or clarifies the request.
3. A router classifies the task.
4. An agent workflow performs the task.
5. A final formatter presents the answer.

Exercise 7.1: Define the use case

Step-by-step instructions

1. Each student selects one project theme.
2. Write the problem statement in two sentences.
3. Identify the target user.
4. Identify the input, processing logic, and output.
5. Define success criteria for the application.

Expected outcome

Students should have a clearly scoped application idea.

Exercise 7.2: Draw the architecture

Step-by-step instructions

1. Draw a block diagram for the system.
2. Include user input, AI generation, routing, agent logic, memory or retrieval if required, and final output.
3. Mark which parts are local and which are API-based.
4. Label data formats between stages, such as plain text, JSON, or table rows.
5. Add one sentence for each block explaining its purpose.

Expected outcome

Students should understand the full pipeline before implementation.

Exercise 7.3: Define prompts and interfaces

Step-by-step instructions

1. Write one system prompt for the generation step.
2. Write one classification prompt for routing.
3. Write one task prompt for the agent.
4. Define the exact output format for each stage.

5. Create sample input-output examples.

Expected outcome

Students should know how the different AI components communicate with each other.

Deliverable for Session 7

- One architecture diagram.
- One page of requirements.
- One prompt set for generation, routing, and agent tasks.

Session 8

Implementation of integrated workflow

Session objective

Students should build the integrated application in which generation and agent steps work together as one system.

Concept coverage

- Chaining AI components
- Input normalization and structured output
- Use of local Llama for agent tasks
- Use of API model for higher-quality generation when needed
- Error handling and fallback logic

Instructor demonstration

The instructor should build a reference workflow such as:

- Chat trigger or form input.
- Request normalizer.
- Generation step to improve the query.
- Router to classify intent.
- Local Llama agent to answer or act.
- Response formatter.
- Optional logging step.

Exercise 8.1: Build the input layer

Step-by-step instructions

1. Create the entry point in n8n using a trigger node.

2. Capture user input from a chat form or webhook.
3. Validate that the input is not empty.
4. Convert the input into a standard object with fields such as ``user_text``, ``topic``, and ``priority``.

Expected outcome

Students should have a reliable application input layer.

Exercise 8.2: Add a generation step

Step-by-step instructions

1. Use ChatGPT or a Groq-backed step to refine the user request.
2. Ask it to clean up the text, infer missing context, or generate a better prompt.
3. Pass the improved text to the next step.
4. Compare original input and refined input.

Expected outcome

Students should understand how a generator can prepare work for downstream automation.

Exercise 8.3: Add agent routing and action

Step-by-step instructions

1. Add a classifier node or branching logic.
2. Route the request into one of at least three paths: explain, summarize, or extract.
3. Use the local Llama model for the agent response.
4. Add a final formatting step so the output is readable to the user.

Expected outcome

Students should build a working agentic system, not just a single prompt call.

Exercise 8.4: Add persistence or logging

Step-by-step instructions

1. Store the input, route, and final output in a local file, sheet, or database.
2. Include timestamp and task type.
3. Use the log to trace errors and response quality later.

Expected outcome

Students should understand the importance of traceability in AI applications.

Deliverable for Session 8

- One working integrated workflow.
- Screenshot of the workflow graph.
- One exported workflow file.
- One sample log of successful runs.

Session 9

Testing, refinement, and demo

Session objective

Students should evaluate their integrated AI application, improve weak points, and present a polished demo.

Concept coverage

- Test cases and edge cases
- Prompt failure modes
- Consistency and repeatability
- User experience and response formatting
- Demo storytelling

Instructor demonstration

The instructor should show how to test:

- A normal request.
- A messy or ambiguous request.
- An unsupported request.
- A request with missing context.

Exercise 9.1: Test with multiple scenarios

Step-by-step instructions

1. Create at least five test inputs.
2. Include simple, complex, ambiguous, and invalid examples.
3. Run each input through the workflow.
4. Record whether each path behaves correctly.
5. Note failures and anomalies.

Expected outcome

Students should learn to test AI systems systematically.

Exercise 9.2: Improve weak points

Step-by-step instructions

1. Find one issue in the workflow, such as poor routing or weak formatting.
2. Modify the prompt, route, or output format.
3. Rerun the failing test.
4. Compare before and after results.

Expected outcome

Students should understand that integrated AI systems usually need iteration.

Exercise 9.3: Create the demo script

Step-by-step instructions

1. Write a 2-minute demo explanation.
2. Show the problem statement.
3. Show the architecture.
4. Show one successful live run.
5. Explain what makes the system useful and where its limits are.

Expected outcome

Students should be able to communicate the application clearly to a technical audience.

Exercise 9.4: Final presentation

Step-by-step instructions

1. Present the application to the class.
2. Demonstrate one sample input and output.
3. Explain the design choices.
4. Mention the local versus cloud tradeoff.
5. Answer peer questions.

Expected outcome

Students should complete the lab with confidence in their integrated AI application.

Assessment plan

Component	Weightage
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Architecture and planning	20%
Workflow implementation	30%
Integration quality	20%
Testing and refinement	20%
Demo and explanation	10%

Evaluation rubric

Criterion	Excellent	Good	Needs improvement
System design	Clear end-to-end architecture with logical stages	Mostly clear with minor gaps	Weak or incomplete architecture
Integration	Components work together reliably	Workflow works with occasional issues	Components are not connected properly
Reliability	Good handling of edge cases and formatting	Some test coverage	Little testing or error handling
Usability	Output is clear and useful to end users	Output is acceptable but uneven	Output is hard to use
Presentation	Explains design and tradeoffs well	Adequate presentation	Unclear or incomplete demo

Expected cumulative outcomes

By the end of Sessions 7-9, students should be able to:

- Design a real AI application that uses both generation and agentic automation.
- Build a working end-to-end system using local and API-based components.
- Test, refine, and explain the behavior of the application.
- Present a complete prototype suitable for evaluation.

Suggested instructor notes

- Encourage students to keep the architecture simple enough to finish within the lab window.
- Use one shared rubric so students know what “good” looks like.
- Remind students that the best AI systems are usually structured systems, not just large prompts.
- Push students to write down failure cases; this makes the final reflection much stronger.

Materials to provide students

- Architecture template.
- Prompt template sheet.
- Workflow export starter file.
- Test case worksheet.
- Demo checklist.

Submission checklist

Students must submit:

- Architecture diagram.
- Prompt set.
- Workflow export.
- Test case results.
- Final demo notes.

Expected final lab outcome

At the end of this third lab block, students should have a complete AI application that combines generative AI and local agent workflows into one coherent system, demonstrating both conceptual understanding and practical implementation ability.

